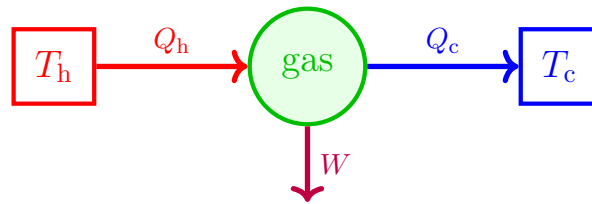


Modelling & Simulation: Assignment 4 (option 3)

Heat engines, heat pumps and thermodynamic cycles

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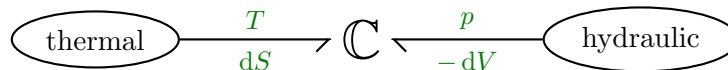
Introduction



Made famous by scientists and engineers like Carnot, Otto and Stirling; driver of automobiles, refrigerators and heat pumps; bane of thermodynamics pupils, *perpetuum mobiles* and Modelling & Simulation students—your assignment this week is to get to grips with the **heat engine**, or more specifically, with the **ideal thermodynamic cycles** that are at their core.

Assignment

Model and simulate two different heat engine cycles¹ using an ideal working gas.



Your models should include (at least):

- a multi-port $\mathbb{C}$ -element modelling an ideal working gas, with
 - a mechanical (hydraulic, (p, dV)) port and
 - a thermal (true thermal, (T, dS)) port; and
- sources and control algorithms to implement two ideal thermodynamic cycles.¹

Some aspects you should focus on in your simulation experiments are listed below.

- Do the port variables show the expected characteristics of the cycle process?
- How does the efficiency of your processes compare to the theoretic maximum?
- Does your heat engine show expected behaviour when changing parameters like (high and low) temperature, “stroke size” (ΔV), cycle time, et cetera?

¹Ideal thermodynamic cycles usually consist of 4 phases, of which at least one compression and one expansion, and involving heating and cooling of the working gas. Examples include Carnot, Stirling, Otto, Diesel, Brayton, Ericsson, and Atkinson.